

Wireshark

Wireshark is a free and open source packet analyser. It is a computer program that can intercept and log traffic passing over a digital network or part of a network. As data flows across the path, the sniffer captures each packet and, if needed, decodes its raw data, showing the values of various fields in the packet and analyses its content. Packet capture is the process of intercepting and logging traffic.

Wireshark runs on GNU/Linux, OS X, BSD, Solaris, some other Unix-like operating systems and Microsoft Windows. There is also a terminal based (non-GUI) version called TShark. Wireshark and other programs distributed with it such as TShark are free software, released under the terms of the GNU general-public licence. Network types supported on various platforms by Wireshark are shown in Fig. 1.

Wireshark is very similar to tcpdump (common packet analyser that runs under command line) but has a graphical front-end, plus some integrated sorting and filtering options.

Features of Wireshark

Data can be captured 'from the wire' from a live network connection, or read from a file of already-captured packets. Live data can be read from a number of types of network including Ethernet, IEEE 802.11 and PPP. Captured network data can be browsed via a GUI or via the terminal (command line) version of the utility, TShark. Captured files can be programmatically edited or converted via command-line switches to editcap program. Data display can be refined using a display filter.

Plugins can be created for dissecting new protocols. Voice over Internet Protocol, or VoIP, calls in the captured traffic can be detected. If encoded in compatible encoding, media flow can even be played. Raw USB traffic can be captured, packets on many criteria filtered/

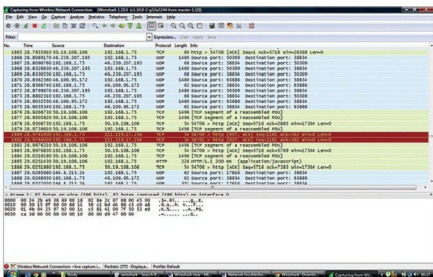


Fig. 3: Live packet capture in action using Wireshark

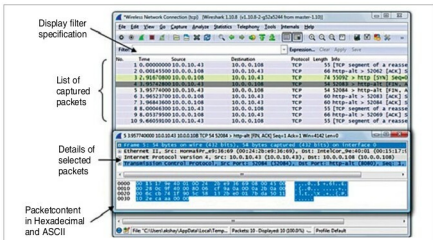


Fig. 4: Different components on Wireshark window

searched and captured packet data saved.

Getting started with Wireshark

To get started with Wireshark, follow the steps given below:

1. Wireshark can be downloaded from its official website link www.wireshark.org/download.html
2. Follow the instructions for installation.
3. Run the application.
4. Select the appropriate network hardware connection you wish to check or trace (for example, local area network or wireless network connection)
5. Click Start on Home screen

(Fig. 2) to start capturing live packets from the hardware connection specified.

Fig. 4 demonstrates display filter specification and the list of packets being captured by Wireshark. This information is used for data analysis and for finding a probable error in the communication system.

Details of different Wireshark components and operations are given below:

1. On the left-most side of Wireshark window (Fig. 4) we have two parameters: packet number and time. Both parameters are very important to decide the sequence and time of packet captured. For example, we